



Ataribox retro mini-console

Atari made a new mini-console inspired by the 1977 classic Atari 2600, for both retro games and modern titles. <http://digit.in/Ataribox1>



AI in a USB stick

California-based Movidius' new Neural Compute Stick is a little thumb drive with a machine vision processor inside. <http://digit.in/USBai>

Are graphics card boxes finally here to stay?

G

raphics card docks aren't a new innovation at all. They've been around for a lot longer than most

companies would like to admit while they're busy portraying their own devices as "the world's first". In fact, some of the earlier docks used to have proprietary connectors which in part was due to laptop manufacturers coming up with proprietary expansion I/O interfaces. However, the proprietary nature of these docks wasn't



Alienware Graphics Amplifier

great for the consumer as they were quite expensive owing to the rarity of these devices. And with Internet services not being so fast in those days, (we're talking of more than a decade back) there was no urgency to deliver compute intensive content faster. Not only were services like 3D rendering and CAD application expensive but they also needed a server farm to ensure a fast turn around time. These proprietary boxes would make the process only ever so slightly faster so investing in them was only for those with deep pockets.

A RESURGENCE OF GPU DOCKS

Come 2016 and we suddenly saw a surge in these devices. And the credit for this surge goes to Intel, NVIDIA, AMD, and certain third-party vendors. Intel, because of its Thunderbolt standard, finally led to enough bandwidth being made available to make external graphics card docks feasible and away from the clutches of proprietary interfaces. NVIDIA, because its Maxwell and Pascal architectures were very power efficient. This made it possible to power these external graphics card boxes with small power adapters. AMD's Polaris architecture is also power efficient but NVIDIA



Razer Core

External graphics card docks give small form factor machines a shot in the arm but with GPU advancements, their days could be numbered.

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ASUS ROG XG Station 2

still has more powerful graphics cards. Laptops can and do house low-end and mid-range GPUs, so the demand for external graphics card docks was driven by the high-end power efficient cards which only NVIDIA has had for the last couple of years. AMD's hero cards have been quite notorious for their high power consumption. VEGA might change that perception for the better. And lastly, huge credit goes to third-party vendors like Razer which jumped onto the external dock bandwagon and provided some much needed competition.

Despite of all these factors coming into play, we still didn't see as many